

Warspear Online (MMORPG, RPG)

com.aigrind.warspear

Developer	TITULUM
Genre	GAME_ROLE_PLAYING
Installs	5,000,000+
Audit Date	2026-04-26

Overall Risk Score

0.37

MEDIUM

6

Consistent

3

Mismatches

11

Over-disclosure

3

Undeclared

Research prototype · Ongoing research | Policy: <http://warspear-online.com/aa/privacy>

Executive Summary

Warspear Online (MMORPG, RPG) declares 6 data type(s) collected and 2 shared in its Play Data Safety label, across 23 data type(s) analyzed. Our heterogeneous GNN audit model identifies **3 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **3 policy-vs-label mismatch(es)**. Overall risk score: **0.37 (MEDIUM)**.

Top discrepancies by risk:

- **installed_apps**: Undeclared Collection (confidence 0.97)
- **diagnostics**: Undeclared Collection (confidence 0.96)
- **approximate_location**: Undeclared Collection (confidence 0.63)

Class definitions:

CONSISTENT: Label, policy, and SDK signals agree.

POLICY_LABEL_MISMATCH: Label discloses data not mentioned in policy.

OVER_DISCLOSURE: Policy mentions data not declared in label.

UNDECLARED_COLLECTION: SDK signals collection undisclosed in label and policy.

Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

Data Type	Class	Conf.	Evidence
installed_apps	Undeclared Collection	0.97	label:N policy:N sdk:Y (1, 2, 3 +5)
diagnostics	Undeclared Collection	0.96	label:N policy:N sdk:Y (1, 2, 3 +5)
approximate_location	Undeclared Collection	0.63	label:N policy:N sdk:Y (1, 2, 3 +5)
crash_logs	Policy Label Mismatch	0.99	label:Y policy:N sdk:Y (1, 2, 3 +5)
purchase_history	Policy Label Mismatch	0.99	label:Y policy:N sdk:N (1, 2, 3 +5)
emails	Policy Label Mismatch	0.98	label:Y policy:N sdk:N (1, 2, 3 +5)
contacts	Over Disclosure	0.96	label:N policy:Y sdk:N (1, 2, 3 +5)
other_user_generated_content	Over Disclosure	0.96	label:N policy:Y sdk:N (1, 2, 3 +5)
user_payment_info	Over Disclosure	0.95	label:N policy:Y sdk:N (1, 2, 3 +5)
files_and_docs	Over Disclosure	0.95	label:N policy:Y sdk:N (1, 2, 3 +5)
precise_location	Over Disclosure	0.95	label:N policy:Y sdk:N (1, 2, 3 +5)
date_of_birth	Over Disclosure	0.93	label:N policy:Y sdk:N (1, 2, 3 +5)
health_info	Over Disclosure	0.91	label:N policy:Y sdk:N (1, 2, 3 +5)
phone_number	Over Disclosure	0.90	label:N policy:Y sdk:N (1, 2, 3 +5)
calendar_events	Over Disclosure	0.90	label:N policy:Y sdk:N (1, 2, 3 +5)
other_info	Over Disclosure	0.81	label:N policy:Y sdk:N (1, 2, 3 +5)
fitness_info	Over Disclosure	0.76	label:N policy:Y sdk:N (1, 2, 3 +5)
device_id	Consistent	1.00	label:Y policy:Y sdk:Y (1, 2, 3 +5)
user_ids	Consistent	1.00	label:Y policy:Y sdk:N (1, 2, 3 +5)
email_address	Consistent	1.00	label:N policy:Y sdk:Y (1, 2, 3 +5)
name	Consistent	1.00	label:N policy:Y sdk:Y (1, 2, 3 +5)
advertising_id	Consistent	0.99	label:N policy:Y sdk:Y (1, 2, 3 +5)
app_interactions	Consistent	0.99	label:Y policy:Y sdk:N (1, 2, 3 +5)

Evidence Trails

Detailed evidence for UNDECLARED_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

Discrepancy: installed_apps — Undeclared Collection (confidence 0.97)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Add 'installed_apps' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: diagnostics — Undeclared Collection (confidence 0.96)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Add 'diagnostics' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: approximate_location — Undeclared Collection (confidence 0.63)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Add 'approximate_location' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: crash_logs — Policy Label Mismatch (confidence 0.99)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'crash_logs' collection, aligning it with the Play Data Safety label.

Discrepancy: purchase_history — Policy Label Mismatch (confidence 0.99)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'purchase_history' collection, aligning it with the Play Data Safety label.

Discrepancy: emails — Policy Label Mismatch (confidence 0.98)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'emails' collection, aligning it with the Play Data Safety label.

Discrepancy: contacts — Over Disclosure (confidence 0.96)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'contacts' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: other_user_generated_content — Over Disclosure (confidence 0.96)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'other_user_generated_content' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: user_payment_info — Over Disclosure (confidence 0.95)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'user_payment_info' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: files_and_docs — Over Disclosure (confidence 0.95)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'files_and_docs' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: precise_location — Over Disclosure (confidence 0.95)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'precise_location' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: date_of_birth — Over Disclosure (confidence 0.93)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'date_of_birth' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: health_info — Over Disclosure (confidence 0.91)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'health_info' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: phone_number — Over Disclosure (confidence 0.90)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'phone_number' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: calendar_events — Over Disclosure (confidence 0.90)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Review whether 'calendar_events' is genuinely collected. If yes, add to label; if no, remove from policy.

Methodology

PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md.

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

Disclaimer

This report is produced by a research prototype and is provided for informational and research purposes only. It does not constitute legal advice or a regulatory determination. Findings are based on automated weak-supervision labels from publicly available data and have not been reviewed by a privacy legal professional. Research prototype · Ongoing research.